

PATHFINDER.

BY PATHFINDER LIDAR SOLUTIONS

PRODUCT STRATEGY

Editions & Integrations

The open-core model (Community Edition and Pathfinder Pro) and third-party services.

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This document defines the open-core product model (a free open-source edition plus a paid commercial edition with paying users from day one) and the third-party services the platform uses or should adopt.

1. Open-core model

Pathfinder ships in two editions from a single codebase lineage:

- **Pathfinder Community Edition (CE)** - open source (AGPL-3.0), self-hostable, single-brand. A genuinely useful floorplan review + safety-mapping portal a school district or fire department could run themselves.
- **Pathfinder Pro** - commercial, hosted SaaS operated by Pathfinder LiDAR Solutions, with paying customers from launch. Pro adds the white-label/multi-tenant, 3D scan hosting, collaboration-at-scale, compliance, and enterprise-auth capabilities that the research identifies as the monetizable differentiators.

Pathfinder LiDAR owns the copyright and therefore dual-licenses: CE under AGPL-3.0 (so anyone running a modified version as a network service must publish their changes), and Pro under a commercial license (see `COMMERCIAL.md`). This protects the hosted-SaaS moat while keeping the core honestly open.

1.1 Why this split

The competitive research (research/01) shows the market is full of heavy bundled platforms. CE earns trust, adoption, and contributions with a real tool. Pro monetizes exactly the capabilities that are expensive to operate (3D splat hosting, large-file storage, compliance exports, multi-tenant white-label) and that institutional buyers will pay for - the same wedge incumbents leave open.

2. Edition feature matrix

Mapped to the canonical epics (E1-E14). "Core" = a usable subset; "Full" = the complete capability.

Capability (epic)	Community Edition	Pathfinder Pro
White-label brand layer (E1)	Single brand (self-host your own)	Full multi-brand / drop-in operator branding; resell under any brand
Identity, roles & access (E2)	JWT + roles (admin/staff/client), audit log	+ SSO/SAML/OIDC, MFA, SCIM provisioning
Org hierarchy & dashboards (E3)	Single org; basic project list	District > Facility > Building hierarchy + roll-up dashboards
Project & review workflow (E4)	Full	Full + approvals/SLA reporting
2D annotation engine (E5)	Full (ported, all tools)	Full
Safety mapping & NFPA export (E6)	NFPA markers + legend + single-map PDF export	+ Batch export pipeline, templated multi-building runs
Unified scan library & media (E7)	Local/self-managed storage, basic versioning	Managed R2 hosting, multipart upload, cold archive, quotas
3D scan viewer (E8)	mp4 walkthrough playback	

Capability (epic)	Community Edition	Pathfinder Pro
		+ Interactive Gaussian-splat viewer, 3D measurement, anchored 3D markers, floor switching, viewpoints, version compare
Collaboration (E9)	In-app comment threads + resolve	+ @mentions, batched email notifications, external share links
Global search (E10)	Per-project search	Cross-org FTS5 global search
Compliance & trust (E11)	Audit log (in-app)	NG911/NENA GeoJSON export, immutable audit export, compliance metadata, trust page, DPA
Accessibility (E12)	WCAG 2.1 AA UI	WCAG 2.1 AA UI + maintained VPAT/ACR
Admin & platform ops (E13)	User + API-key management	+ Multi-tenant admin, usage metering, billing
Quality & delivery harness (E14)	Full (open test suite)	Full
Billing & subscriptions	n/a (self-host)	Stripe-based subscriptions, metered storage
Support	Community (issues/discussions)	Priority support + SLA

3. Third-party services & integrations

3.1 In use / core stack

Service	Role	Edition	Notes
Cloudflare (Pages, Workers, D1, R2, KV)	Hosting, DB, object storage, cache	Both	Core platform; CE can self-host on a free CF account
Resend	Transactional email (invites, password reset, notifications, @mentions)	Both (CE optional)	v1 already wired RESEND_API_KEY / RESEND_FROM

3.2 Adopt now (priority - required for Pro launch)

Service	Role	Why now
Stripe	Subscriptions, Checkout, Customer Portal, usage/ metered billing via webhooks (Workers)	Pro has paying users from day one - billing is launch-blocking
Cloudflare Turnstile	Bot/abuse protection on login + public forms	Free, native to the stack, low effort
Sentry (or Cloudflare Observability + Logpush)	Error monitoring, performance, release health	Production reliability for paying customers

3.3 Forward / partner integrations (Pro, market-driven)

Service	Role	Trigger
RapidSOS	Push verified maps/location data to PSAPs (NG911 clearinghouse)	The universal 911-buyer integration (research/01); pursue partner program

Service	Role	Trigger
Geocoding (Mapbox or Google Maps)	Facility address -> coordinates for NG911/ NENA GIS export and map context	When NG911 GeoJSON export (E11) ships
SSO / Identity (WorkOS, or self-hosted OIDC/SAML)	Enterprise SSO + MFA + SCIM	First enterprise gov/edu customer requiring SSO
SIS integration (Clever / ClassLink)	School roster sync for reunification/account provisioning	K-12 reunification feature work
Antivirus / content scanning	Scan uploaded PDFs/media before serving	Before opening uploads to untrusted client users at scale

3.4 Offline pipeline (not runtime APIs)

Tool	Role
FJD Trion Model	Point cloud -> Gaussian splat generation + mp4 walkthrough export (desktop)
PlayCanvas SplatTransform (CLI)	Convert PLY -> SPZ (web delivery format)
SuperSplat editor	Clean floaters/outliers before delivery

3.5 Secrets these imply

All set as Cloudflare secrets / env vars, never committed (see `AGENTS.md`): `JWT_SECRET` , `RESEND_API_KEY` , `STRIPE_SECRET_KEY` , `STRIPE_WEBHOOK_SECRET` , `TURNSTILE_SECRET_KEY` , `SENTRY_DSN` , and (forward) `RAPIDSOS_*` , `MAPBOX_TOKEN` , SSO client secrets.

4. Decisions & open questions

- **Decision:** CE = AGPL-3.0; Pro = commercial dual-license. Single repo public; Pro-only modules may live behind a feature flag or a separate private package - to be finalized in S1.
- **Decision:** Stripe is the billing provider for Pro.
- **Open:** Whether Pro-only code ships in the public repo behind flags, or in a private overlay repo. Recommendation: keep CE complete and honest in public; gate Pro features by license/flag, with server-enforced entitlement checks.
- **Open:** RapidSOS and SSO vendor selection deferred until first customer demand.